

# Experimental Noise Literature Database

## Superconducting qubits and surface-code quantum error correction

This database records experimentally supported noise mechanisms, their timescales, detector signatures, interactions, interventions, and source citations. Missing values are preserved as not reported. Measurements from different devices are not combined into one hardware model.

## Evidence labels

Directly measured: reported experimental measurement. Fitted: obtained from a fit. Derived from same-device measurements: calculated only from compatible measurements in the same experiment. Theoretical/model parameter: used in a model or protocol. Not reported: no defensible value located in the cited material.

# Main mechanism records

Each mechanism is shown as a field/value record so the wide database remains readable in PDF form.

## T1 energy relaxation

| Field                          | Recorded information                                                                                                                                                                                                                                                 |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device/platform                | Google Willow, 105-qubit superconducting transmon processor                                                                                                                                                                                                          |
| Physical cause                 | Energy loss from the ket(1) state to ket(0) through electromagnetic/material/environmental channels; TLS defects can modulate relaxation                                                                                                                             |
| Error probability/rate         | Not reported as a single per-cycle probability in the main article                                                                                                                                                                                                   |
| Timescale type                 | T1 relaxation time: population decay constant for excited-state energy relaxation                                                                                                                                                                                    |
| Timescale value                | Mean T1 = 68 microseconds                                                                                                                                                                                                                                            |
| Uncertainty                    | Mean only in main text; distribution/uncertainty not given there                                                                                                                                                                                                     |
| Decay/correlation model        | T1 is normally extracted from exponential excited-state decay; the article does not give the fit equation in the accessible main text                                                                                                                                |
| Model parameters               | T1 = 68 microseconds, directly measured/fitted device summary                                                                                                                                                                                                        |
| QEC cycle time                 | 1.1 microseconds                                                                                                                                                                                                                                                     |
| Persistence in QEC rounds      | Not reported. T1 is not leakage lifetime or detector-correlation time                                                                                                                                                                                                |
| Qubit type                     | Data qubits and measure qubits in the Willow surface-code architecture                                                                                                                                                                                               |
| Circuit location               | Idle periods and every operation through decoherence exposure                                                                                                                                                                                                        |
| Spatial extent                 | Primarily single qubit; TLS/environment can make the relaxation fluctuate locally                                                                                                                                                                                    |
| Temporal signature             | Baseline relaxation is approximately memoryless at the operation-model level; T1 itself can drift in time                                                                                                                                                            |
| Detector/syndrome signature    | Can contribute isolated detection events and time-like effects; detailed detector signature not separately assigned in the main text                                                                                                                                 |
| Interaction with other noise   | Changes operation fidelity and can coexist with leakage, dephasing, and calibration drift                                                                                                                                                                            |
| Intervention/mitigation result | Higher coherence from fabrication, participation-ratio engineering, and circuit optimization                                                                                                                                                                         |
| Measurement type               | T1 calibration on physical qubits; surface-code memory experiment                                                                                                                                                                                                    |
| Source                         | Google Quantum AI and Collaborators, "Quantum error correction below the surface code threshold," Nature 638, 920-926 (2025), DOI 10.1038/s41586-024-08449-y ( <a href="https://doi.org/10.1038/s41586-024-08449-y">https://doi.org/10.1038/s41586-024-08449-y</a> ) |
| Figure/table/section           | Main text, lines describing processor coherence; Fig. 1b component-error distributions                                                                                                                                                                               |
| Evidence classification        | Directly measured device summary; fitted definition of T1                                                                                                                                                                                                            |
| Notes                          | Do not use 68 microseconds as tau_c. It is an energy-relaxation decay constant.                                                                                                                                                                                      |

T2 dephasing

| Field                          | Recorded information                                                                                                                                                                       |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device/platform                | Google Willow, 105-qubit superconducting transmon processor                                                                                                                                |
| Physical cause                 | Loss of phase coherence from frequency/flux fluctuations, thermal photons, TLS and control/environmental noise                                                                             |
| Error probability/rate         | Not reported as a single per-cycle probability in the main article                                                                                                                         |
| Timescale type                 | T2,CPMG: coherence decay time under a CPMG dynamical-decoupling sequence; not the same as Ramsey T2* or a noise correlation time                                                           |
| Timescale value                | Mean T2,CPMG = 89 microseconds                                                                                                                                                             |
| Uncertainty                    | Mean only in main text; distribution/uncertainty not given there                                                                                                                           |
| Decay/correlation model        | The main text does not state the fitted decay equation for this summary value                                                                                                              |
| Model parameters               | T2,CPMG = 89 microseconds, directly measured/fitted device summary                                                                                                                         |
| QEC cycle time                 | 1.1 microseconds                                                                                                                                                                           |
| Persistence in QEC rounds      | Not reported                                                                                                                                                                               |
| Qubit type                     | Data and measure transmons                                                                                                                                                                 |
| Circuit location               | Idle and gate intervals; phase-sensitive operations                                                                                                                                        |
| Spatial extent                 | Primarily single-qubit, with possible spatially correlated control/environment contributions                                                                                               |
| Temporal signature             | Can include fast dephasing plus low-frequency components; the article does not reduce it to one correlation time                                                                           |
| Detector/syndrome signature    | Phase faults appear in detector patterns associated with Z-type faults; detailed attribution is not isolated in the main article                                                           |
| Interaction with other noise   | Low-frequency noise can also cause coherent gate errors and calibration drift                                                                                                              |
| Intervention/mitigation result | CPMG/dynamical-decoupling coherence measurement; hardware and circuit optimization                                                                                                         |
| Measurement type               | T2 calibration and surface-code component-error benchmarking                                                                                                                               |
| Source                         | Google Quantum AI and Collaborators, Nature (2025), DOI 10.1038/s41586-024-08449-y ( <a href="https://doi.org/10.1038/s41586-024-08449-y">https://doi.org/10.1038/s41586-024-08449-y</a> ) |
| Figure/table/section           | Main text, coherence paragraph; Fig. 1b                                                                                                                                                    |
| Evidence classification        | Directly measured device summary; fitted definition of T2,CPMG                                                                                                                             |
| Notes                          | T2,CPMG suppresses selected low-frequency noise and must not be treated as a universal exponential correlation lifetime.                                                                   |

Single-qubit gate errors

| Field                          | Recorded information                                                                                                                                                                                                                                                       |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device/platform                | Fixed-frequency tantalum-film transmon, Q5, five-qubit device                                                                                                                                                                                                              |
| Physical cause                 | Pulse-amplitude/phase/detuning error, control-line distortion, decoherence during the pulse, leakage and non-Markovian coherent error                                                                                                                                      |
| Error probability/rate         | Average EPG = $(7.42 \pm 0.04) \times 10^{-5}$ ; decoherence lower bound = $(4.62 \pm 0.04) \times 10^{-5}$ ; leakage rate per Clifford = $(2.57 \pm 0.08) \times 10^{-5}$ ; abstract also reports leakage per gate = $(1.16 \pm 0.04) \times 10^{-5}$                     |
| Timescale type                 | Gate duration and RB sequence-decay scale; no physical memory time reported                                                                                                                                                                                                |
| Timescale value                | Gate length = 20 ns; maximum RB sequence = 4,500 Clifford gates                                                                                                                                                                                                            |
| Uncertainty                    | EPG uncertainty $\pm 0.04 \times 10^{-5}$ ; decoherence lower bound $\pm 0.04 \times 10^{-5}$ ; leakage $\pm 0.04 \times 10^{-5}$ per gate and $\pm 0.08 \times 10^{-5}$ per Clifford                                                                                      |
| Decay/correlation model        | RB: $F_{\text{seq}}(m) = A p^m + B$ ; purity: $P_{\text{seq}}(m) = A u^{(m-1)} + B$ ; leakage population: $P_{2,m} = P_{\text{inf}}(1 - \exp(-\Gamma_{\text{clif}} m)) + P_0 \exp(-\Gamma_{\text{clif}} m)$                                                                |
| Model parameters               | 2.2083 $\pi/2$ pulses per Clifford; $d=2$ ; $T_1 = 231$ microseconds; $T_2^E = 204$ microseconds                                                                                                                                                                           |
| QEC cycle time                 | Not a QEC-cycle experiment                                                                                                                                                                                                                                                 |
| Persistence in QEC rounds      | Not reported                                                                                                                                                                                                                                                               |
| Qubit type                     | Fixed-frequency transmon                                                                                                                                                                                                                                                   |
| Circuit location               | Single-qubit X/Y $\pi/2$ pulses and virtual-Z decomposition                                                                                                                                                                                                                |
| Spatial extent                 | Single target qubit; control-line/circuit effects can spread through simultaneous operation                                                                                                                                                                                |
| Temporal signature             | RB error is modeled as exponential in sequence length, but the paper reports non-Markovian behavior in long-sequence GST                                                                                                                                                   |
| Detector/syndrome signature    | Not reported                                                                                                                                                                                                                                                               |
| Interaction with other noise   | Coherent residual errors can accumulate nonlocally in QEC circuits; leakage creates non-Pauli effects                                                                                                                                                                      |
| Intervention/mitigation result | Alternating reference/interleaved RB to reduce drift; pulse calibration and GST; paper identifies non-Markovianity as bottleneck                                                                                                                                           |
| Measurement type               | Interleaved randomized benchmarking, purity benchmarking, gate-set tomography                                                                                                                                                                                              |
| Source                         | Zhiyuan Li et al., "Error per single-qubit gate below $10^{-4}$ in a superconducting qubit," npj Quantum Information 9, 111 (2023), DOI 10.1038/s41534-023-00781-x ( <a href="https://doi.org/10.1038/s41534-023-00781-x">https://doi.org/10.1038/s41534-023-00781-x</a> ) |
| Figure/table/section           | Fig. 1; Results sections on RB and purity benchmarking; abstract                                                                                                                                                                                                           |
| Evidence classification        | Directly measured and fitted                                                                                                                                                                                                                                               |
| Notes                          | The listed EPG is an average gate metric, not a probability that remains correlated for a specified time.                                                                                                                                                                  |

Two-qubit gate errors

| Field                          | Recorded information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device/platform                | Sycamore/Google superconducting transmon processors; surface-code experiments use diabatic CZ gates                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Physical cause                 | Imperfect entangling pulse, residual ZZ/stray interactions, decoherence during CZ, leakage and coherent phase errors                                                                                                                                                                                                                                                                                                                                                                                                               |
| Error probability/rate         | General transmon range stated as below 10^-2 in Li et al.; Google Willow component distributions are shown in Fig. 1b but a scalar range is not given in the accessible main text                                                                                                                                                                                                                                                                                                                                                  |
| Timescale type                 | Gate duration; no standalone temporal correlation time reported                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Timescale value                | Sub-100 ns CZ is described in crosstalk analysis; exact surface-code CZ duration is not reported in the accessible sources reviewed                                                                                                                                                                                                                                                                                                                                                                                                |
| Uncertainty                    | Not reported for a universal range                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Decay/correlation model        | Surface-code detector-event pair probabilities and correlated matching graphs rather than one decay law                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Model parameters               | Error model contains Pauli CZ errors, leakage, stray interactions, and detection hyperedges                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| QEC cycle time                 | Willow surface-code cycle = 1.1 microseconds; Krinner distance-3 cycle = 1.1 microseconds                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Persistence in QEC rounds      | Not reported as a single persistence count                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Qubit type                     | Data-data, data-measure, and measure-neighbor transmons                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Circuit location               | CZ/entangling layers; also idle intervals while other qubits are measured/reset                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Spatial extent                 | Nearest-neighbor and spectator-qubit effects; can form multi-detector clusters                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Temporal signature             | Mostly short gate-local faults, but leakage and stray interactions produce longer spatial/temporal correlations                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Detector/syndrome signature    | Spacelike and spacetime detector pairs; unexpected distant pairs attributed/surmised to leakage and stray interactions                                                                                                                                                                                                                                                                                                                                                                                                             |
| Interaction with other noise   | Strongly interacts with leakage, crosstalk, idle errors, and Y-type correlated X/Z effects                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Intervention/mitigation result | Correlated matching synthesis, leakage removal, pulse/coupler tuning, and device design                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Measurement type               | Two-qubit RB/XEB and surface-code detector-pair correlation analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Source                         | Google, “Suppressing quantum errors by scaling a surface code logical qubit,” Nature 614, 676-681 (2023), DOI 10.1038/s41586-022-05434-1 ( <a href="https://doi.org/10.1038/s41586-022-05434-1">https://doi.org/10.1038/s41586-022-05434-1</a> ); Google, Nature (2025); Sundaresan et al., “Quantum Crosstalk Analysis for Simultaneous Gate Operations,” PRX Quantum 3, 020301 (2022), DOI 10.1103/PRXQuantum.3.020301 ( <a href="https://doi.org/10.1103/PRXQuantum.3.020301">https://doi.org/10.1103/PRXQuantum.3.020301</a> ) |
| Figure/table/section           | Nature 2023 Fig. 2 and discussion of detector pair probabilities; PRX Quantum abstract/results                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Evidence classification        | Directly measured detector correlations plus model components; exact gate-error distributions are source-specific                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Notes                          | Do not replace correlated detector-pair data with a single independent two-qubit error number.                                                                                                                                                                                                                                                                                                                                                                                                                                     |

Measurement/readout errors

| Field                          | Recorded information                                                                                                                                                                                                                                                  |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device/platform                | Superconducting transmons; examples include Chalmers transmon readout and Google surface-code processors                                                                                                                                                              |
| Physical cause                 | Assignment overlap, state relaxation during readout, amplifier/SNR limitations, state-preparation and measurement (SPAM) effects                                                                                                                                      |
| Error probability/rate         | Two-state assignment fidelity 99.5% => assignment infidelity approximately 0.5%; three-state fidelity 96.9%; during 140 ns readout, ket(1) decay contribution epsilon = 2.24%                                                                                         |
| Timescale type                 | Readout pulse duration and relaxation during readout                                                                                                                                                                                                                  |
| Timescale value                | Readout time = 140 ns; T01 = 6.18 microseconds for the measured qubit                                                                                                                                                                                                 |
| Uncertainty                    | Not reported                                                                                                                                                                                                                                                          |
| Decay/correlation model        | Exponential relaxation contribution epsilon = 1 - exp(-tau_r/T01); multilevel readout uses sums of exponentials                                                                                                                                                       |
| Model parameters               | Effective T01 after shelving = 245 microseconds; standard T01 = 6.18 microseconds; 80 repetitions x 50,000 shots for assignment data                                                                                                                                  |
| QEC cycle time                 | Surface-code cycle from Krinner = 1.1 microseconds; Chalmers experiment is not a surface-code cycle                                                                                                                                                                   |
| Persistence in QEC rounds      | Not reported                                                                                                                                                                                                                                                          |
| Qubit type                     | Ancilla/measure qubits and final data-qubit readout                                                                                                                                                                                                                   |
| Circuit location               | Measurement, readout pulse, and final logical readout                                                                                                                                                                                                                 |
| Spatial extent                 | Single channel or multiplexed resonator group; readout crosstalk can affect several qubits                                                                                                                                                                            |
| Temporal signature             | Shot-to-shot approximately stochastic; relaxation during the pulse is time-dependent and multilevel decay is non-single-exponential                                                                                                                                   |
| Detector/syndrome signature    | False detector events; wrong syndrome bits; measurement errors are identification errors in Willow Fig. 1b                                                                                                                                                            |
| Interaction with other noise   | Readout error couples to T1 during measurement, leakage, resonator crosstalk and SPAM                                                                                                                                                                                 |
| Intervention/mitigation result | Shelving to higher levels increased effective T01; machine-learning post-processing; readout calibration and assignment-matrix correction                                                                                                                             |
| Measurement type               | Single-shot IQ histograms and multilevel population-vs-time measurements                                                                                                                                                                                              |
| Source                         | Liangyu Chen et al., "Transmon qubit readout fidelity at the threshold for quantum error correction without a quantum-limited amplifier," arXiv:2208.05879 (2022), DOI/link arXiv ( <a href="https://arxiv.org/abs/2208.05879">https://arxiv.org/abs/2208.05879</a> ) |
| Figure/table/section           | Fig. 2 and Fig. 3; Results sections II.1-II.3                                                                                                                                                                                                                         |
| Evidence classification        | Directly measured/fitted                                                                                                                                                                                                                                              |
| Notes                          | The 0.5% value is inferred from the reported 99.5% fidelity and should be retained as derived from that same experiment.                                                                                                                                              |

Reset/preparation errors

| Field                          | Recorded information                                                                                                                                                                                                                                                           |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device/platform                | Sycamore flux-tunable transmons with readout resonators; multi-level reset experiment                                                                                                                                                                                          |
| Physical cause                 | Incomplete ground-state preparation, failed swaps, diabatic return, residual excitation and readout floor                                                                                                                                                                      |
| Error probability/rate         | Reset error floor about 0.2%; dominant modeled error channel about $5 \times 10^{-4}$ ; second channel much less than $10^{-4}$ ; theory deviation below $10^{-3}$                                                                                                             |
| Timescale type                 | Reset completion and resonator decay times                                                                                                                                                                                                                                     |
| Timescale value                | Reset protocol about 250 ns; resonator decay time $1/\kappa = 45$ ns; hold about 300 ns                                                                                                                                                                                        |
| Uncertainty                    | Not reported as a statistical uncertainty for the floor in the main text                                                                                                                                                                                                       |
| Decay/correlation model        | Reset error decreases exponentially with hold time until readout floor; model uses Landau-Zener and resonator decay                                                                                                                                                            |
| Model parameters               | $t_{\text{swap}} = 30$ ns; $\Delta_f = 2.5$ GHz; $g$ about 120 MHz; $P_D(r)$ about 0.6; qubit idle frequency 6.09 GHz; resonator 4.67 GHz; hold detuning 1 GHz; nonlinearity -200 MHz                                                                                          |
| QEC cycle time                 | Bit-flip code round = 955 ns including reset; not a 2D surface-code cycle                                                                                                                                                                                                      |
| Persistence in QEC rounds      | Leakage growth/removal tracked over code rounds, but a single reset-error persistence count is not reported                                                                                                                                                                    |
| Qubit type                     | Measure qubits directly; data-qubit leakage can be transported to measure qubits through CZ                                                                                                                                                                                    |
| Circuit location               | Reset after measurement; leakage removal on measure qubits and, in later protocols, data qubits                                                                                                                                                                                |
| Spatial extent                 | Primarily single qubit; transport can move leakage between neighboring data/measure qubits                                                                                                                                                                                     |
| Temporal signature             | Short-lived reset dynamics; robust to parameter drift in the reported operating region                                                                                                                                                                                         |
| Detector/syndrome signature    | Reduces leakage-induced correlated detector-event tails and correlations                                                                                                                                                                                                       |
| Interaction with other noise   | Reset directly changes leakage lifetime; readout floor limits measured reset error; CZ transports leakage from data to measure qubits                                                                                                                                          |
| Intervention/mitigation result | Multi-level reset; 40x increase in $\gamma_{\text{down}}$ for measure qubits and 2.4x increase for data qubits in the bit-flip-code experiment                                                                                                                                 |
| Measurement type               | State-resolved population measurement, reset-error landscape, bit-flip-code leakage growth                                                                                                                                                                                     |
| Source                         | Google, "Removing leakage-induced correlated errors in superconducting quantum error correction," Nature Communications 12, 697 (2021), DOI 10.1038/s41467-021-21982-y ( <a href="https://doi.org/10.1038/s41467-021-21982-y">https://doi.org/10.1038/s41467-021-21982-y</a> ) |
| Figure/table/section           | Fig. 2 and Results reset-gate section; Fig. 3 and leakage section                                                                                                                                                                                                              |
| Evidence classification        | Directly measured plus fitted/model parameters                                                                                                                                                                                                                                 |
| Notes                          | Reset is not just a preparation detail: it controls whether leakage becomes a multi-round correlated error.                                                                                                                                                                    |

Leakage

| Field                          | Recorded information                                                                                                                                                                                                                                             |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device/platform                | Google Sycamore transmons in distance-3 surface code and distance-21 bit-flip code                                                                                                                                                                               |
| Physical cause                 | Population leaves computational ket(0),ket(1) space into ket(2) and higher levels, especially during entangling gates and measurement/reset                                                                                                                      |
| Error probability/rate         | Average device leakage population below 1 x 10^-3 after removal; tenfold reduction in steady-state data leakage; injected initial population near 50%                                                                                                            |
| Timescale type                 | Leakage decay/persistence time in QEC cycles                                                                                                                                                                                                                     |
| Timescale value                | Injected leakage exponential decay constant about 4.4 surface-code cycles; each cycle approximately 1 microsecond                                                                                                                                                |
| Uncertainty                    | “Around 4.4 cycles”; uncertainty not reported in main text                                                                                                                                                                                                       |
| Decay/correlation model        | Excess leakage population decays exponentially; expected isolated T1 decay shown separately; leakage also spatially spreads                                                                                                                                      |
| Model parameters               | Initial injected ket(2) population near 50%; circuit uses diabatic CZ resonance; no single universal leakage-generation probability reported                                                                                                                     |
| QEC cycle time                 | Approximately 1 microsecond per cycle                                                                                                                                                                                                                            |
| Persistence in QEC rounds      | Directly reported: persists for many rounds; decay constant about 4.4 cycles; not converted into a hard cutoff                                                                                                                                                   |
| Qubit type                     | Data and measure qubits                                                                                                                                                                                                                                          |
| Circuit location               | During CZ/entangling operations, measurement, reset, and higher-level transitions                                                                                                                                                                                |
| Spatial extent                 | Injected on one central data qubit, then transported to neighbors and eventually all qubits involved at d=3                                                                                                                                                      |
| Temporal signature             | Long-lived across many cycles and spatially transported; explicitly correlated                                                                                                                                                                                   |
| Detector/syndrome signature    | Pair of detection events at injection plus a tail of correlated detection events over leakage lifetime; increased event fraction across affected qubits                                                                                                          |
| Interaction with other noise   | Strong interaction with CZ leakage transport, correlated Pauli errors, measurement, and reset                                                                                                                                                                    |
| Intervention/mitigation result | Data-qubit and measure-qubit leakage removal; tenfold steady-state reduction and stable QEC performance                                                                                                                                                          |
| Measurement type               | State-resolved leakage measurement and detector-event fraction after deliberate ket(1) to ket(2) injection                                                                                                                                                       |
| Source                         | Google Quantum AI and Collaborators, “Overcoming leakage in quantum error correction,” Nature Physics 20, 875-881 (2024), DOI 10.1038/s41567-023-02226-w ( <a href="https://doi.org/10.1038/s41567-023-02226-w">https://doi.org/10.1038/s41567-023-02226-w</a> ) |
| Figure/table/section           | Fig. 1b-c, text around leakage dynamics; Fig. 2 leakage transport                                                                                                                                                                                                |
| Evidence classification        | Directly measured and fitted                                                                                                                                                                                                                                     |
| Notes                          | This is the clearest directly reported QEC-round persistence value in the database.                                                                                                                                                                              |

Crosstalk and stray interactions

| Field                          | Recorded information                                                                                                                                                                                                                                                                                  |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device/platform                | Fixed-frequency/tunable-bus transmon architectures; Google surface-code processors                                                                                                                                                                                                                    |
| Physical cause                 | Microwave cross-driving, flux/control-line coupling, residual ZZ, parasitic spectator interactions and frequency collisions                                                                                                                                                                           |
| Error probability/rate         | Exact universal measured probability is not reported in the crosstalk-analysis paper; PRX Quantum theory identifies “considerable” leakage/control error near parasitic resonances                                                                                                                    |
| Timescale type                 | Interaction duration and gate-local coherent accumulation                                                                                                                                                                                                                                             |
| Timescale value                | Sub-100 ns CZ gate context; sub-MHz parasitic next-nearest-neighbor interactions                                                                                                                                                                                                                      |
| Uncertainty                    | Not reported                                                                                                                                                                                                                                                                                          |
| Decay/correlation model        | Coherent/non-Pauli spectator errors; no single decay law                                                                                                                                                                                                                                              |
| Model parameters               | Coupling-induced dressing; parasitic next-neighbor interactions; collision-region dependence                                                                                                                                                                                                          |
| QEC cycle time                 | Surface-code cycle 1.1 microseconds in Google experiments                                                                                                                                                                                                                                             |
| Persistence in QEC rounds      | Not reported                                                                                                                                                                                                                                                                                          |
| Qubit type                     | Data, measure and spectator qubits                                                                                                                                                                                                                                                                    |
| Circuit location               | Simultaneous single-qubit microwave drives, flux pulses, CZ/CZ spectator periods                                                                                                                                                                                                                      |
| Spatial extent                 | Nearest neighbors and next-nearest-neighbor spectators; potentially multiple qubits                                                                                                                                                                                                                   |
| Temporal signature             | Explicitly spatially correlated; may be coherent and repeatable rather than memoryless                                                                                                                                                                                                                |
| Detector/syndrome signature    | Unexpected spatially separated detector pairs and higher-weight clusters; Google 2023 attributes unexpected pairs in part to leakage and stray interactions                                                                                                                                           |
| Interaction with other noise   | Interacts with two-qubit gate errors, leakage, calibration and frequency placement                                                                                                                                                                                                                    |
| Intervention/mitigation result | Frequency/coupler design, pulse scheduling, calibration and avoiding collision regions; exact lifetime change not reported                                                                                                                                                                            |
| Measurement type               | Simultaneous-operation benchmarking and theoretical crosstalk analysis; surface-code detector-pair statistics                                                                                                                                                                                         |
| Source                         | M. A. H. et al., “Quantum Crosstalk Analysis for Simultaneous Gate Operations on Superconducting Qubits,” PRX Quantum 3, 020301 (2022), DOI 10.1103/PRXQuantum.3.020301 ( <a href="https://doi.org/10.1103/PRXQuantum.3.020301">https://doi.org/10.1103/PRXQuantum.3.020301</a> ); Google Nature 2023 |
| Figure/table/section           | PRX Quantum abstract/results; Google Nature 2023 Fig. 2 discussion                                                                                                                                                                                                                                    |
| Evidence classification        | Directly observed/theoretical depending on value                                                                                                                                                                                                                                                      |
| Notes                          | Preserve this as a mechanism class until a device-specific crosstalk matrix is available.                                                                                                                                                                                                             |

Low-frequency / 1/f noise

| Field                          | Recorded information                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device/platform                | MIT/RIKEN/Waterloo superconducting persistent-current flux qubit                                                                                                                                                                                                                                                                                                                                   |
| Physical cause                 | Fluctuating magnetic flux and tunneling amplitude; likely environmental/material fluctuators                                                                                                                                                                                                                                                                                                       |
| Error probability/rate         | PSD measured, not a single probability; $1/f^\alpha$ with $\alpha$ about 0.9                                                                                                                                                                                                                                                                                                                       |
| Timescale type                 | Spectral band, quasi-static cutoff and dephasing time                                                                                                                                                                                                                                                                                                                                              |
| Timescale value                | 0.2-20 MHz noise spectroscopy; same power law consistent with 0.01-100 Hz; typical $\tau_0 = 1$ microsecond; $T_{\phi^*} = 3.2$ microseconds in one flux-qubit setting                                                                                                                                                                                                                             |
| Uncertainty                    | $\alpha$ uncertainty not given in accessible abstract; $T_{\phi^*}$ value reported without uncertainty                                                                                                                                                                                                                                                                                             |
| Decay/correlation model        | $S(f)$ proportional to $1/f^\alpha$ ; for strict $1/f$ , Ramsey/free-induction decay is Gaussian rather than exponential                                                                                                                                                                                                                                                                           |
| Model parameters               | $\alpha$ about 0.9; $T_1 = 12$ microseconds; acquisition time 10 s; temperature 65-200 mK; source reports $T_{\phi^*} = (2 \pi \kappa_\Delta D_\Delta)^{-1} A_\Delta^{-1/2} [\ln(t_{\text{acq}}/(2 \tau_0))]^{-1/2}$                                                                                                                                                                               |
| QEC cycle time                 | Not a surface-code experiment                                                                                                                                                                                                                                                                                                                                                                      |
| Persistence in QEC rounds      | Not reported                                                                                                                                                                                                                                                                                                                                                                                       |
| Qubit type                     | Persistent-current flux qubit, not transmon; relevance is mechanism-level                                                                                                                                                                                                                                                                                                                          |
| Circuit location               | Idle/free evolution and driven dynamics; dynamical-decoupling filter function                                                                                                                                                                                                                                                                                                                      |
| Spatial extent                 | Primarily one qubit in the measurement; cross-PSD between two inferred noise sources was measured                                                                                                                                                                                                                                                                                                  |
| Temporal signature             | Long-lived low-frequency/quasi-static component; explicitly colored and non-Markovian                                                                                                                                                                                                                                                                                                              |
| Detector/syndrome signature    | Dephasing changes phase-flip statistics; surface-code detector signature not measured in this device                                                                                                                                                                                                                                                                                               |
| Interaction with other noise   | Multiple noise sources were tested for cross-correlation; no evidence of time-domain correlation between the two inferred noise sources in the 2012 study                                                                                                                                                                                                                                          |
| Intervention/mitigation result | Dynamical decoupling and Rabi spectroscopy characterize/filter the spectrum; suppression changes observed coherence but the source does not report a QEC-round lifetime                                                                                                                                                                                                                            |
| Measurement type               | Ramsey/free-induction, dynamical decoupling, Rabi spectroscopy, cross-PSD                                                                                                                                                                                                                                                                                                                          |
| Source                         | Jonas Bylander et al., “Noise spectroscopy through dynamical decoupling with a superconducting flux qubit,” Nature Physics 7, 565-570 (2011), DOI 10.1038/nphys1994 ( <a href="https://doi.org/10.1038/nphys1994">https://doi.org/10.1038/nphys1994</a> ); Fei Yan et al., “Spectroscopy of low-frequency noise and its temperature dependence in a superconducting qubit,” arXiv:1201.5665 (2012) |
| Figure/table/section           | Bylander p. 565 and Fig. 2; Yan Fig. 2-4, section III.2.2                                                                                                                                                                                                                                                                                                                                          |
| Evidence classification        | Directly measured PSD; fitted $\alpha$ and $T_{\phi^*}$ ; model equation                                                                                                                                                                                                                                                                                                                           |
| Notes                          | A $1/f$ spectrum does not have one finite correlation time without specifying low/high-frequency cutoffs.                                                                                                                                                                                                                                                                                          |

Calibration drift / fluctuating device parameters

| Field                          | Recorded information                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device/platform                | Google transmons with TLS probes; 12-qubit flux-tunable transmon preprint for sub-second dynamics                                                                                                                                                                                                                                                                         |
| Physical cause                 | TLS resonance motion, environmental changes, pulse/control drift and fluctuating qubit frequency/relaxation                                                                                                                                                                                                                                                               |
| Error probability/rate         | T1 can fluctuate by up to an order of magnitude in time/frequency in the Klimov paper; no single drift probability                                                                                                                                                                                                                                                        |
| Timescale type                 | Drift timescale / recalibration cadence                                                                                                                                                                                                                                                                                                                                   |
| Timescale value                | Klimov: fluctuations are unpredictable in frequency/time; precise universal tau not reported. A 2026 12-transmon preprint shows T1 fluctuations over a 2-second acquisition window and motivates sub-second calibration                                                                                                                                                   |
| Uncertainty                    | Not reported as a universal uncertainty                                                                                                                                                                                                                                                                                                                                   |
| Decay/correlation model        | Drift is nonstationary and may be multi-timescale; not a single exponential in the cited summaries                                                                                                                                                                                                                                                                        |
| Model parameters               | 2026 preprint: T1 estimate 10 ms, readout optimization 100 ms, pulse optimization 1 ms, Clifford RB 107 ms; 74,000+ recalibrations over 6 h                                                                                                                                                                                                                               |
| QEC cycle time                 | Not a QEC-cycle measurement; Willow QEC cycle 1.1 microseconds for context only                                                                                                                                                                                                                                                                                           |
| Persistence in QEC rounds      | Not reported                                                                                                                                                                                                                                                                                                                                                              |
| Qubit type                     | Data and measure qubits; calibration parameters affect both                                                                                                                                                                                                                                                                                                               |
| Circuit location               | Gate amplitudes, frequencies, readout thresholds, idle frequencies and schedules                                                                                                                                                                                                                                                                                          |
| Spatial extent                 | Local TLS effects plus chip-level control/environment channels                                                                                                                                                                                                                                                                                                            |
| Temporal signature             | Slowly drifting, intermittent or abruptly changing; nonstationary                                                                                                                                                                                                                                                                                                         |
| Detector/syndrome signature    | Changing detector rates over time; long-lived correlated blocks are possible, but exact surface-code detector autocorrelation not given in the cited calibration papers                                                                                                                                                                                                   |
| Interaction with other noise   | Drift changes effective T1/T2, gate error, leakage and readout error; calibration can mask or respond to the drift                                                                                                                                                                                                                                                        |
| Intervention/mitigation result | Recalibration, TLS-aware design, FPGA closed-loop calibration; the 2026 preprint reports better-than-baseline gate errors during continuous recalibration                                                                                                                                                                                                                 |
| Measurement type               | Repeated T1, readout, pulse-amplitude and Clifford-RB measurements                                                                                                                                                                                                                                                                                                        |
| Source                         | P. V. Klimov et al., "Fluctuations of Energy-Relaxation Times in Superconducting Qubits," PRL 121, 090502 (2018), DOI 10.1103/PhysRevLett.121.090502 ( <a href="https://doi.org/10.1103/PhysRevLett.121.090502">https://doi.org/10.1103/PhysRevLett.121.090502</a> ); "Millisecond-Scale Calibration and Benchmarking of Superconducting Qubits," arXiv:2602.11912 (2026) |
| Figure/table/section           | Klimov abstract; 2026 preprint Fig. 1 and abstract                                                                                                                                                                                                                                                                                                                        |
| Evidence classification        | Directly measured fluctuations; fitted/model timing parameters where stated                                                                                                                                                                                                                                                                                               |
| Notes                          | Calibration cadence is an intervention time, not automatically the physical correlation time.                                                                                                                                                                                                                                                                             |

Correlated multi-qubit / high-energy events

| Field                          | Recorded information                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device/platform                | Modular superconducting processor with motherboard and two flip-chip daughterboard modules, each module containing two qubits; Google high-distance repetition/surface-code processors                                                                                                                                                                                                                   |
| Physical cause                 | Cosmic-ray-like or other high-energy events temporarily suppress relaxation across several qubits; other chip-wide correlated faults remain partly unexplained                                                                                                                                                                                                                                           |
| Error probability/rate         | Within-module decay-event correlation >85%; physically separated modules about 2%; Google repetition-code floor $1.7 \times 10^{-6}$ logical error/cycle, $1.6 \times 10^{-7}$ excluding one event                                                                                                                                                                                                       |
| Timescale type                 | Burst correlation and event-rate timescale                                                                                                                                                                                                                                                                                                                                                               |
| Timescale value                | Exact burst duration not reported in the accessible arXiv abstract; Google 2025 reports approximately one event per hour or $3 \times 10^9$ cycles for rare correlated events                                                                                                                                                                                                                            |
| Uncertainty                    | >85% and about 2% reported without uncertainty in abstract; Google logical-floor values reported without uncertainty in the cited summary                                                                                                                                                                                                                                                                |
| Decay/correlation model        | Bursty, correlated, non-Poisson-looking at event level; temporary suppression of T1; no universal exponential fit reported                                                                                                                                                                                                                                                                               |
| Model parameters               | One event/hour; $3 \times 10^9$ cycles; $1.7 \times 10^{-6}$ and $1.6 \times 10^{-7}$ per cycle; spatial correlations >85% vs about 2%                                                                                                                                                                                                                                                                   |
| QEC cycle time                 | Google surface-code cycle 1.1 microseconds                                                                                                                                                                                                                                                                                                                                                               |
| Persistence in QEC rounds      | Google reports event frequency, not number of rounds per burst; do not derive rounds from the hourly rate                                                                                                                                                                                                                                                                                                |
| Qubit type                     | Multiple data/ancilla qubits and modules                                                                                                                                                                                                                                                                                                                                                                 |
| Circuit location               | Can occur during any circuit operation; observed as decay-event bursts                                                                                                                                                                                                                                                                                                                                   |
| Spatial extent                 | Strong within a module; much weaker across separated modules; Google reports chip-scale rare events                                                                                                                                                                                                                                                                                                      |
| Temporal signature             | Rare, bursty, spatially correlated, long-tail contribution to logical performance                                                                                                                                                                                                                                                                                                                        |
| Detector/syndrome signature    | Many simultaneous decay/detection events; creates clusters conventional local decoders may misinterpret                                                                                                                                                                                                                                                                                                  |
| Interaction with other noise   | Coupled to architecture/modularity and possibly quasiparticles/TLS; the source does not establish a single cause                                                                                                                                                                                                                                                                                         |
| Intervention/mitigation result | Modular separation reduces within-system correlation; specialized codes/mitigation are proposed; Google surface-code work uses leakage removal and correlated decoding but does not claim this source is eliminated                                                                                                                                                                                      |
| Measurement type               | Repeated decay monitoring, repetition-code logical memories and correlation analysis                                                                                                                                                                                                                                                                                                                     |
| Source                         | "Mitigating cosmic ray-like correlated events with a modular quantum processor," arXiv:2505.15919 (2025), arXiv link ( <a href="https://arxiv.org/abs/2505.15919">https://arxiv.org/abs/2505.15919</a> ); Google Quantum AI and Collaborators, Nature 638 (2025), DOI 10.1038/s41586-024-08449-y ( <a href="https://doi.org/10.1038/s41586-024-08449-y">https://doi.org/10.1038/s41586-024-08449-y</a> ) |
| Figure/table/section           | Cosmic-ray arXiv abstract; Google Nature abstract and Fig. 3 discussion                                                                                                                                                                                                                                                                                                                                  |
| Evidence classification        | Directly measured                                                                                                                                                                                                                                                                                                                                                                                        |
| Notes                          | This mechanism is deliberately separated from ordinary two-qubit correlated errors because its spatial scale and burst statistics are different.                                                                                                                                                                                                                                                         |

# Coexisting mechanisms and collective temporal behavior

| Experiment                                   | Coexisting mechanisms                                                                                                     | Collective measurement                                                                                                                                                                                 | What changed when one mechanism was controlled                                                                                                                                   | Source and location                                                                  | Evidence classification                                                   |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Google distance-5 surface-code scaling       | Pauli gate errors, idle errors, readout errors, leakage, stray interactions, Y-type X/Z correlations                      | Detection-event probabilities over space/time for 25 cycles and normalized detector-pair correlations $p_{ij}$ ; unexpected pairs were more frequent experimentally than in a local depolarizing model | Adding leakage and stray-interaction structure to the Pauli model brought predictions closer to experiment; leakage removal is identified as increasingly important              | Google, Nature 614 (2023), Fig. 2 and detector-pair discussion                       | Directly measured detector-pair correlations plus fitted model comparison |
| Google distance-3 leakage injection          | Leakage, CZ transport, T1 relaxation, measurement and reset                                                               | Excess leakage population versus cycle and detection-event fraction versus detector location/round                                                                                                     | Removing leakage from all qubits prevented performance decline over time and shortened leakage persistence; injected leakage without removal had decay constant about 4.4 cycles | Google, Nature Physics (2024), Fig. 1c and Fig. 4                                    | Directly measured and fitted model comparison                             |
| Google Willow below-threshold surface code   | Single-qubit Pauli errors, CZ errors, data idle errors, measurement identification errors, leakage and stray interactions | Component error distributions, weight-4 detection probabilities averaged over 250 cycles, logical error versus memory duration                                                                         | Data-qubit leakage removal after syndrome extraction keeps leakage short-lived; correlated matching synthesis and neural decoding improve model agreement/accuracy               | Google, Nature 638 (2025), Fig. 1b-d and main text                                   | Directly measured and fitted model comparison                             |
| Sycamore bit-flip code with multilevel reset | Leakage generation, leakage transport through CZ, reset errors, readout floor, T1 relaxation                              | Leakage population growth/saturation versus stabilizer-code length; fitted $\gamma_{up}$ and $\gamma_{down}$ ; leakage-induced error-pair probabilities                                                | Reset increased $\gamma_{down}$ by 40x for measure qubits and 2.4x for data qubits on average; applying reset reduced correlation magnitude                                      | Google, Nature Communications 12, 697 (2021), Fig. 3 and leakage-correlation results | Directly measured and fitted model comparison                             |
| Flux-qubit noise spectroscopy                | Low-frequency flux noise, tunneling-amplitude noise, white noise and possible cross-correlation                           | PSDs over 0.01-100 Hz and 0.2-20 MHz; cross-correlation/cross-PSD between inferred noise sources                                                                                                       | Dynamical-decoupling filter functions separated frequency bands; no time-domain correlation between the two inferred noise sources was found                                     | Yan et al. arXiv:1201.5665, Fig. 2-4 and Appendix                                    | Directly measured PSD/PSD ratio and fitted model                          |
| High-energy/burst studies                    | Correlated decay events, local T1 suppression, module coupling                                                            | Coincident decay events within and between modules                                                                                                                                                     | Physical separation reduced observed correlation from >85% within a module to about 2% across separated modules                                                                  | arXiv:2505.15919, abstract                                                           | Directly measured                                                         |

# Raw numerical parameter table

Every numerical value is retained separately without averaging across devices.

| ID  | Raw value | Units             | Parameter / definition                                                          | Device / experiment                              | Kind                                         | Exact source and location                                                     |
|-----|-----------|-------------------|---------------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------|-------------------------------------------------------------------------------|
| P01 | 68        | microseconds      | Mean T1, energy-relaxation time                                                 | Google Willow, 105-qubit processor               | Directly measured; fitted definition         | Google Nature 638 (2025), main text coherence paragraph                       |
| P02 | 89        | microseconds      | Mean T2,CPMG, CPMG coherence time                                               | Google Willow, 105-qubit processor               | Directly measured; fitted definition         | Google Nature 638 (2025), main text coherence paragraph                       |
| P03 | 1.1       | microseconds      | Surface-code QEC cycle                                                          | Google Willow and Krinner distance-3 experiment  | Directly measured                            | Google Nature 638 (2025), abstract; Krinner Nature (2022), abstract           |
| P04 | 0.143     | percent per cycle | Distance-7 logical error per cycle, 101-qubit memory                            | Google Willow                                    | Directly measured/fitted                     | Google Nature 638 (2025), abstract/main text                                  |
| P05 | 0.003     | percent           | Uncertainty on P04                                                              | Google Willow                                    | Directly measured/fitted uncertainty         | Google Nature 638 (2025), abstract                                            |
| P06 | 2.14      | dimensionless     | Error suppression factor Lambda for increasing distance by 2                    | Google Willow                                    | Fitted                                       | Google Nature 638 (2025), abstract                                            |
| P07 | 0.02      | dimensionless     | Uncertainty on P06                                                              | Google Willow                                    | Fitted uncertainty                           | Google Nature 638 (2025), abstract                                            |
| P08 | 2.4       | dimensionless     | Logical lifetime / best physical-qubit lifetime                                 | Google Willow                                    | Fitted/derived from same-device measurements | Google Nature 638 (2025), abstract                                            |
| P09 | 0.3       | dimensionless     | Uncertainty on P08                                                              | Google Willow                                    | Fitted uncertainty                           | Google Nature 638 (2025), abstract                                            |
| P10 | 3         | percent per cycle | Logical error probability with leakage-detected runs rejected                   | 17-qubit distance-3 superconducting surface code | Directly measured                            | Krinner et al., Nature (2022), abstract                                       |
| P11 | 17        | physical qubits   | Device size                                                                     | Krinner distance-3 surface code                  | Directly measured                            | Krinner et al., Nature (2022), abstract                                       |
| P12 | 25        | cycles            | Memory duration in Google distance-3/5 scaling comparison                       | Google Sycamore surface-code experiment          | Directly measured protocol parameter         | Google Nature 614 (2023), abstract/main text                                  |
| P13 | 2.914     | percent           | Distance-5 logical error per cycle                                              | Google Sycamore                                  | Directly measured/fitted                     | Google Nature 614 (2023), abstract/main text                                  |
| P14 | 0.016     | percent           | Uncertainty on P13                                                              | Google Sycamore                                  | Fitted uncertainty                           | Google Nature 614 (2023), abstract/main text                                  |
| P15 | 3.028     | percent           | Average distance-3 logical error per cycle                                      | Google Sycamore                                  | Directly measured/fitted                     | Google Nature 614 (2023), abstract/main text                                  |
| P16 | 0.023     | percent           | Uncertainty on P15                                                              | Google Sycamore                                  | Fitted uncertainty                           | Google Nature 614 (2023), abstract/main text                                  |
| P17 | 1.7e-6    | per cycle         | Distance-25 repetition-code logical-error floor including one high-energy event | Google Sycamore                                  | Directly measured                            | Google Nature 614 (2023), abstract                                            |
| P18 | 1.6e-7    | per cycle         | Same floor excluding the high-energy event                                      | Google Sycamore                                  | Directly measured                            | Google Nature 614 (2023), abstract                                            |
| P19 | 4.4       | cycles            | Exponential decay constant of injected leakage population                       | Google Sycamore distance-3 surface code          | Fitted                                       | Overcoming leakage, Nature Physics (2024), Fig. 1c/text                       |
| P20 | 1         | microsecond       | Approximate surface-code cycle during leakage experiment                        | Google Sycamore distance-3 surface code          | Directly measured protocol parameter         | Overcoming leakage, Nature Physics (2024), Fig. 1c caption/text               |
| P21 | 50        | percent           | Near-maximum injected ket(2) population due to random initial ket(1) population | Google Sycamore leakage-injection experiment     | Directly measured/derived protocol bound     | Overcoming leakage, Nature Physics (2024), Fig. 1c/text                       |
| P22 | 1e-3      | population        | Average leakage population after removal, less than this value                  | Google Sycamore surface-code processor           | Directly measured                            | Overcoming leakage, Nature Physics (2024), abstract                           |
| P23 | 10        | fold              | Steady-state data leakage reduction                                             | Google Sycamore surface-code processor           | Directly measured ratio                      | Overcoming leakage, Nature Physics (2024), abstract                           |
| P24 | 250       | nanoseconds       | Multilevel reset duration                                                       | Sycamore transmon reset experiment               | Directly measured protocol parameter         | Nature Communications 12, 697 (2021), Fig. 2/text                             |
| P25 | 0.2       | percent           | Reset/readout visibility floor                                                  | Sycamore transmon reset experiment               | Directly measured                            | Nature Communications 12, 697 (2021), Fig. 2/text                             |
| P26 | 45        | nanoseconds       | Resonator decay time 1/kappa                                                    | Sycamore transmon reset experiment               | Fitted/model parameter                       | Nature Communications 12, 697 (2021), Fig. 2/text                             |
| P27 | 5e-4      | probability       | Dominant modeled reset error channel                                            | Sycamore transmon reset experiment               | Theoretical/model parameter                  | Nature Communications 12, 697 (2021), reset model text                        |
| P28 | <<1e-4    | probability       | Second modeled reset error channel                                              | Sycamore transmon reset experiment               | Theoretical/model bound                      | Nature Communications 12, 697 (2021), reset model text                        |
| P29 | 955       | nanoseconds       | Bit-flip-code round including reset                                             | Sycamore 21-qubit chain                          | Directly measured protocol parameter         | Nature Communications 12, 697 (2021), Fig. 3/text                             |
| P30 | 14        | microseconds      | Average T1 near selected qubits                                                 | Sycamore 21-qubit chain                          | Directly measured                            | Nature Communications 12, 697 (2021), leakage section                         |
| P31 | 40        | states            | Number of random initial data states averaged                                   | Sycamore 21-qubit chain                          | Directly measured protocol parameter         | Nature Communications 12, 697 (2021), leakage section                         |
| P32 | 40        | fold              | Increase in gamma_down for measure qubits with reset                            | Sycamore 21-qubit chain                          | Directly measured ratio                      | Nature Communications 12, 697 (2021), leakage section                         |
| P33 | 2.4       | fold              | Increase in gamma_down for data qubits with reset/transport                     | Sycamore 21-qubit chain                          | Directly measured ratio                      | Nature Communications 12, 697 (2021), leakage section                         |
| P34 | 7.42e-5   | per gate          | Average single-qubit error per gate                                             | Fixed-frequency tantalum transmon Q5             | Directly measured/fitted                     | Li et al., npj Quantum Information 9, 111 (2023), abstract/Results            |
| P35 | 0.04e-5   | per gate          | Uncertainty on P34                                                              | Fixed-frequency tantalum transmon Q5             | Fitted uncertainty                           | Li et al., npj Quantum Information 9, 111 (2023), abstract/Results            |
| P36 | 4.62e-5   | per gate          | Lower bound on decoherence contribution                                         | Fixed-frequency tantalum transmon Q5             | Directly measured lower bound/fitted         | Li et al., npj Quantum Information 9, 111 (2023), abstract                    |
| P37 | 1.16e-5   | per gate          | Leakage rate per gate                                                           | Fixed-frequency tantalum transmon Q5             | Directly measured/fitted                     | Li et al., npj Quantum Information 9, 111 (2023), abstract                    |
| P38 | 2.57e-5   | per Clifford      | Leakage rate per Clifford                                                       | Fixed-frequency tantalum transmon Q5             | Fitted                                       | Li et al., npj Quantum Information 9, 111 (2023), Results purity benchmarking |
| P39 | 0.08e-5   | per Clifford      | Uncertainty on P38                                                              | Fixed-frequency tantalum transmon Q5             | Fitted uncertainty                           | Li et al., npj Quantum Information 9, 111 (2023), Results purity benchmarking |
| P40 | 231       | microseconds      | T1                                                                              | Fixed-frequency tantalum transmon Q5             | Directly measured/fitted                     | Li et al., npj Quantum Information 9, 111 (2023), device description          |
| P41 | 204       | microseconds      | Echo T2^E                                                                       | Fixed-frequency tantalum transmon Q5             | Directly measured/fitted                     | Li et al., npj Quantum Information 9, 111 (2023), device description          |

| ID  | Raw value | Units            | Parameter / definition                                                  | Device / experiment                             | Kind                                 | Exact source and location                                                       |
|-----|-----------|------------------|-------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------|---------------------------------------------------------------------------------|
| P42 | 20        | nanoseconds      | Single-qubit gate length                                                | Fixed-frequency tantalum transmon Q5            | Directly measured protocol parameter | Li et al., npj Quantum Information 9, 111 (2023), device description            |
| P43 | 0.5       | percent          | Approximate two-state assignment infidelity derived from 99.5% fidelity | Chalmers transmon readout                       | Derived from same-device measurement | Chen et al., arXiv:2208.05879, abstract/Fig. 3                                  |
| P44 | 96.9      | percent          | Three-state assignment fidelity                                         | Chalmers transmon readout                       | Directly measured                    | Chen et al., arXiv:2208.05879, abstract                                         |
| P45 | 140       | nanoseconds      | Readout pulse time                                                      | Chalmers transmon readout                       | Directly measured protocol parameter | Chen et al., arXiv:2208.05879, abstract/Results                                 |
| P46 | 2.24      | percent          | ket(1) relaxation contribution during readout                           | Chalmers transmon readout                       | Fitted/derived from same-device T01  | Chen et al., arXiv:2208.05879, Results II.2                                     |
| P47 | 6.18      | microseconds     | T01 used in P46                                                         | Chalmers transmon readout                       | Directly measured/fitted             | Chen et al., arXiv:2208.05879, Results II.2                                     |
| P48 | 245       | microseconds     | Effective T01 with shelving, equivalent readout-decay timescale         | Chalmers transmon readout                       | Derived/model-equivalent             | Chen et al., arXiv:2208.05879, Results II.2                                     |
| P49 | 80        | repetitions      | Assignment-fidelity repetition count                                    | Chalmers transmon readout                       | Directly measured protocol parameter | Chen et al., arXiv:2208.05879, Fig. 3 text                                      |
| P50 | 50000     | shots/repetition | Shots per assignment repetition                                         | Chalmers transmon readout                       | Directly measured protocol parameter | Chen et al., arXiv:2208.05879, Fig. 3 text                                      |
| P51 | 0.9       | exponent         | 1/f^alpha spectral exponent alpha                                       | MIT/RIKEN superconducting flux qubit            | Fitted                               | Bylander et al., Nature Physics 7 (2011), p. 565 / abstract                     |
| P52 | 0.2-20    | MHz              | Dynamical-decoupling noise-spectroscopy band                            | MIT/RIKEN flux qubit                            | Directly measured                    | Bylander et al., Nature Physics 7 (2011), p. 565                                |
| P53 | 0.01-100  | Hz               | Low-frequency noise band measured by Ramsey/free-induction methods      | MIT/RIKEN flux qubit                            | Directly measured                    | Yan et al., arXiv:1201.5665, abstract/text                                      |
| P54 | 5.4-21    | GHz              | High-frequency PSD band inferred from relaxation                        | MIT/RIKEN flux qubit                            | Directly measured/inferred           | Bylander et al., Nature Physics 7 (2011), p. 565                                |
| P55 | 1         | microsecond      | Typical tau_0 cutoff/time parameter in T_phi* analysis                  | MIT/RIKEN flux qubit                            | Theoretical/model parameter          | Yan et al., arXiv:1201.5665, section III.2.2                                    |
| P56 | 10        | seconds          | Acquisition time in T_phi* analysis                                     | MIT/RIKEN flux qubit                            | Directly measured protocol parameter | Yan et al., arXiv:1201.5665, section III.2.2                                    |
| P57 | 12        | microseconds     | T1 used in flux-qubit example                                           | MIT/RIKEN flux qubit                            | Directly measured/fitted             | Yan et al., arXiv:1201.5665, section III.2.2                                    |
| P58 | 3.2       | microseconds     | T_phi* from 1/f dephasing analysis                                      | MIT/RIKEN flux qubit                            | Fitted/derived from same-device data | Yan et al., arXiv:1201.5665, section III.2.2                                    |
| P59 | 65-200    | mK               | Temperature range tested                                                | MIT/RIKEN flux qubit                            | Directly measured protocol range     | Yan et al., arXiv:1201.5665, abstract/text                                      |
| P60 | 10        | milliseconds     | T1-estimation time in sub-second calibration preprint                   | 12-qubit flux-tunable transmon                  | Directly measured protocol parameter | "Millisecond-Scale Calibration and Benchmarking...", arXiv:2602.11912, abstract |
| P61 | 100       | milliseconds     | Readout-optimization time                                               | 12-qubit flux-tunable transmon                  | Directly measured protocol parameter | arXiv:2602.11912, abstract                                                      |
| P62 | 1         | millisecond      | Pulse-amplitude optimization time                                       | 12-qubit flux-tunable transmon                  | Directly measured protocol parameter | arXiv:2602.11912, abstract                                                      |
| P63 | 107       | milliseconds     | Clifford RB time                                                        | 12-qubit flux-tunable transmon                  | Directly measured protocol parameter | arXiv:2602.11912, abstract                                                      |
| P64 | 6         | hours            | Continuous recalibration runtime                                        | 12-qubit flux-tunable transmon                  | Directly measured protocol parameter | arXiv:2602.11912, abstract                                                      |
| P65 | 74000     | recalibrations   | Consecutive recalibrations during 6-hour run                            | 12-qubit flux-tunable transmon                  | Directly measured protocol parameter | arXiv:2602.11912, abstract                                                      |
| P66 | >85       | percent          | Within-module correlated decay events                                   | Modular superconducting processor               | Directly measured                    | arXiv:2505.15919, abstract                                                      |
| P67 | 2         | percent          | Correlation across physically separated modules                         | Modular superconducting processor               | Directly measured                    | arXiv:2505.15919, abstract                                                      |
| P68 | 1         | hour/event       | Approximate rare correlated-event frequency                             | Google high-distance repetition-code experiment | Directly measured                    | Google Nature 638 (2025), abstract                                              |
| P69 | 3e9       | cycles/event     | Same rare-event frequency expressed in cycles                           | Google high-distance repetition-code experiment | Directly measured                    | Google Nature 638 (2025), abstract                                              |